

PROJECT DEVELOPMENT PHASE

Script-Controlled ACL – Restrict Record Access Based on Field Value

The project development phase implements record-level security in ServiceNow using Access Control Lists (ACLs), custom roles, a custom Institution Details table, a Branch field condition, and a script-controlled READ ACL. The development follows the requirements and configuration described in the supplied project document.

1. Development Environment and Requirements

- ServiceNow instance with administrator access.
- Security_admin role for ACL configuration.
- Test user: EEE User.
- Custom roles: bb1, bb2, bb3 and bb4.
- Custom table: u_institution_details.
- Branch values: ECE, EEE and CSE.

2. User and Role Development

A test user named EEE User is created through User Administration → Users. Four custom roles, bb1, bb2, bb3 and bb4, are created under User Administration → Roles. These roles are assigned to the test user and are used to control different record operations.

3. Custom Table Development

The custom table Institution Details is created with the table name u_institution_details. The table stores student-related information and includes a Branch choice field. The Branch field contains ECE, EEE and CSE values.

Field	Type
Student Roll Number	Auto Number
Student Name	Reference - User
Faculty Name	Reference - User
Branch	Choice (ECE, EEE, CSE)
Email	String
Phone Number	String
Description	Multi String

4. Record Creation

Multiple records are created in the Institution Details table with different Branch values such as ECE, EEE and CSE. These records are used to test whether the configured ACLs correctly restrict access.

5. ACL Development

Four record-level ACLs are developed for the u_institution_details table. Each ACL controls a specific operation: READ, CREATE, WRITE and DELETE.

Operation	Required Role	Condition / Purpose
READ	bb1	Branch is EEE; advanced script-controlled ACL
CREATE	bb2	Controls creation of new records
WRITE	bb3	Controls modification of existing records
DELETE	bb4	Controls deletion of existing records

6. READ ACL Development

The READ ACL is configured as an advanced record-level ACL for u_institution_details. The read operation is enabled, bb1 is added as the required role, and the data condition is Branch is EEE. The script allows administrators full access, allows users having bb1 to pass the ACL, and denies access to other users.

READ ACL Script Logic

The script first checks whether the current user has the admin role. If so, access is allowed. It then checks for the bb1 role. If the role is present, access is allowed; otherwise access is denied.

7. CREATE ACL Development

A record-level CREATE ACL is configured for u_institution_details. The required role is bb2 and no data condition is added. This ACL determines whether a user can create a new record. During verification, a user with bb1 and bb2 can see the permitted records and the New button is available.

8. WRITE ACL Development

A record-level WRITE ACL is configured for u_institution_details with bb3 as the required role. No data condition is added. This permission allows a user with the required roles to modify existing permitted records.

9. DELETE ACL Development

A record-level DELETE ACL is configured for u_institution_details with bb4 as the required role. No data condition is added. This ACL controls whether an existing record can be deleted.

10. Testing and Verification

- Impersonate the EEE user and open the Student Records list.
- Verify that the user with bb1 can view the permitted EEE branch records.
- Verify that a user without the required role cannot view the records.
- Verify that an administrator can view all records.
- Verify that users with bb1 and bb2 can create a new record.
- Verify that users with bb1, bb2 and bb3 can edit permitted records.
- Verify that users with bb1, bb2, bb3 and bb4 can view, create, edit and delete permitted EEE branch records.

11. Development Outcome

The development phase results in a ServiceNow security configuration where user roles and record field values are used together to control access. READ, CREATE, WRITE and DELETE operations are separately protected through ACL configuration.

12. Result

The Script-Controlled ACL project is successfully developed by configuring READ, CREATE, WRITE and DELETE access controls for the u_institution_details table. The configured roles and script-based conditions control access to the records.

13. Conclusion

This development demonstrates practical ServiceNow security and ACL configuration. It shows how different user roles can receive different permissions and how a script can be combined with a field condition to control access to specific records. The completed configuration helps protect data from unauthorized viewing, modification, creation and deletion.